

Deployable Self-Regulating Cars: Decentralized Traffic Control with Ordinary Vehicles

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Abstract

Highway traffic jams are a congestion-control failure in a transportation system: small local disturbances push traffic past a critical density, after which stop-and-go waves persist long after their cause has disappeared. Field experiments show that a handful of controlled vehicles can damp these waves, and a recent self-regulating-cars system improves throughput by 5% and cuts average delay by 13% on a real highway network in high-fidelity simulation, yet deployed designs still place purpose-built vehicles, roadside infrastructure, or a cloud coordinator inside the control loop. We argue the remaining obstacle is deployment, and ask two questions: how little must change in an ordinary car for it to self-regulate, and how few self-regulating cars does a road network need? We describe a vehicle that senses traffic with a dashcam-grade camera and a GPS or OBD-II speed feed, maintains an ego-centric digital twin on roughly \$250 of embedded compute, and independently computes safety-filtered speed and headway advisories, using a low-rate aggregate vehicle-to-vehicle exchange to extend its view. Seen this way, a self-regulating fleet is a communication network laid over the road network: edge nodes deciding locally, a low-rate substrate among them, and a cloud confined to training and updates, so what to compute where, what to share, and how much staleness to tolerate are networking questions. A working prototype grounds the design, running a trained policy end to end in 20–25 ms. Minimal fleets complete the argument: on bottleneck-fed corridors like the Holland Tunnel approaches, about 25 vehicles, the ambient presence of one commercial fleet, can regulate the inflow that decides whether the bottleneck jams. We close with the networking and systems questions this opens.

1 Introduction

Traffic jams on free-flow highways are not only a capacity problem but a distributed stability problem: when local density crosses a critical threshold, small braking events, merge bursts, or overreactions amplify into persistent stop-and-go waves even without a crash or fixed bottleneck [3, 28]. The classical traffic-flow view captures this with the fundamental diagram: flow grows with density until a critical operating point, then collapses as interactions dominate [10, 18, 23]. Keeping density below that critical point while holding speed high keeps the stream on the productive branch and maximizes throughput. Small, well-timed local speed adjustments

suffice, which makes decentralized control a plausible lever on a network-level objective.

Fixed highway controllers are poorly matched to this failure mode: ramp meters, variable speed limits, and traffic-management systems act from fixed infrastructure and coarse network views, valuable where deployed but expensive, unevenly available, and slow relative to local disturbances [1, 20]. The vehicle-side actuator, meanwhile, has been demonstrated twice over. In the field, one controlled vehicle among roughly twenty damps stop-and-go waves on a ring track [27], and CIRCLES placed about one hundred controlled vehicles into live interstate traffic, though with purpose-built cars and a centralized speed planner over cellular links [6]. In simulation, a recent self-regulating-cars system showed the policy can be learned: speed modulation on a real highway network improves throughput by 5%, cuts average delay by 13%, and reduces stops by 3% versus no control, applied only at bottleneck segments during peak inflow, with metrics improving steadily as the compliant fraction grows [2]. But that system observes global segment state and broadcasts recommendations through a cloud service.

The timing makes deployment worth taking seriously now. Connected and automated vehicles are moving from pilots to revenue service, so the programmable fraction of a highway stream is rising rather than hypothetical. The computation to estimate local traffic state is already widely available: robotaxis carry substantial onboard compute, smartphones can run the perception and inference, and even human-driven vehicles increasingly ship with cameras and computers. These vehicles are also already networked: tethered to fleet and telematics backends whose driver-facing advisory rail operates at scale today, and, increasingly, able to talk to one another over connected-vehicle radios [4, 25]. Yet each of these vehicles optimizes only its own trip; the effect each car has on the flow around it is left untapped.

We take the position that the remaining problem is deployment, not algorithms and not radios. The effective control law is coarse: discrete speed and headway bins suffice for a learned policy to improve flow [2, 27]. The radio exists: deployed connected-vehicle standards already broadcast per-vehicle safety state at up to ten hertz [7, 16, 24], and the aggregate summaries our design consumes are smaller and slower, so a low-rate exchange rides comfortably on radios the community has already standardized. We spend the paper on what sits above the substrate: (i) *how little must change*

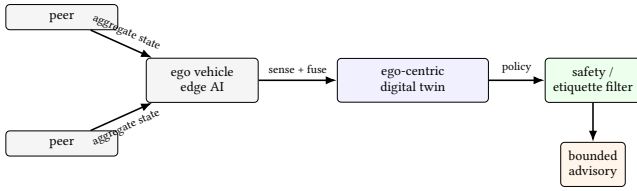


Figure 1: A self-regulating car keeps the real-time loop on board: onboard sensing feeds an ego-centric digital twin, a learned policy proposes a coarse advisory, and a safety and etiquette filter bounds what is shown or executed. Low-rate aggregate summaries from nearby participants, carried over existing radios, extend observability past the camera horizon; no roadside sensor, cloud coordinator, or command exchange is in the loop.

in an existing car for it to become self-regulating, and (ii) how few self-regulating cars must be present, and where, for the network to feel them? Underneath both is a distributed system that networking should recognize: a partially deployed fleet of edge nodes regulating the road network through local decisions, a lossy low-rate substrate among the cars, and a cloud kept off the control path, so where computation lives, what the cars share and how often, and how much staleness the loop can absorb are networking questions we return to in Section 6.

This paper argues that both answers are small, and grounds the argument in a working system. First, it recasts highway congestion control as an edge-deployment problem: ordinary vehicles keep the real-time loop on board, act only through bounded advisories, and treat communication as an observability hint, not a control dependency (Section 2). Second, it gives the bill of changes that makes one car self-regulating: a dashcam-grade camera, an ego-speed feed from GPS or OBD-II, a digital twin and policy that fit on phone-class or \$250 compute, and a safety filter, with no connection to the vehicle’s controls (Section 3). Third, it reports a prototype on a Jetson Orin Nano running a simulation-trained policy against real dashcam footage at a sustained 30 Hz and 20–25 ms end-to-end latency (Section 4). Finally, it argues for a minimal deployment model: because congestion is manufactured at bottlenecks, roughly one vehicle per approach lane, about 25 for a corridor like the Holland Tunnel, could regulate the inflow the bottleneck receives, making a single rideshare or delivery fleet a sufficient launch vehicle (Section 2). Section 6 lays out the open networking questions this system raises.

2 The Case for Self-Regulation

The central claim is not that vehicles should form a fully cooperative swarm. It is that each vehicle should remain the sole decision maker for its own motion, with network-level

behavior emerging from repeated local damping, not from an agreed-upon plan. This distinction matters for deployment. Command-style coordination creates hard questions about authority, liability, and failure: who tells whom to slow down, and what if the message is late? Self-regulation avoids that contract, asking nothing of neighboring vehicles beyond their physical presence.

Why local damping suffices. Highway regulation does not require any vehicle to know the network state, only early evidence that local flow is nearing a bad operating region: density rising near a merge, queues forming downstream, speed variance growing before a shockwave. The control objective is physical and local: keep density below the fundamental diagram’s critical point, then hold speed high and variance low. A vehicle that damps its own speed variance when it sees rising local pressure removes energy from an incipient wave before it steepens, and when enough vehicles do this independently the stream is held near critical density with no shared plan. Even a single uncoordinated vehicle, with no communication at all, already damps stop-and-go waves on ring tracks and open roads [6, 27], so decentralized local action is not the speculative part. The property to protect as actuators are added is string stability: disturbances must not amplify as they propagate upstream through a chain of followers [30], which gives the safety layer and evaluation a concrete target. The analogy to backpressure scheduling, which stabilizes queues from local pressure differences alone, is direct [31]; but packets do not brake hard when startled, which is why Section 3 pairs the controller with a safety filter.

Communication model. Better-than-myopic observability is where communication earns its place, and we keep its role thin. Participating vehicles exchange compact, anonymous, aggregate summaries: nearby counts and density for participants and ordinary traffic, mean speed, lane distribution, a local queue estimate, downstream congestion, and merge pressure. They never exchange raw video, stable identities, acceleration or braking commands, or joint lane plans. Three receiver-side semantics matter more than any message format: expired peers are dropped rather than extrapolated, unheard peers map to explicit neutral values rather than zeros with hidden meaning, and the aggregate is advisory evidence, never a command. Our prototype realizes the substrate as 2 Hz broadcasts with a 150 m radius and a 2 s time-to-live; a summary is tens of bytes, so the footprint is kilobits per second and does not grow with scene complexity. Any deployed radio can carry it, whether DSRC, C-V2X, or in the limit a cellular relay [7, 16, 24]. If no peer is heard, a vehicle degrades to a local-only controller. Because the consumed unit is a neighborhood aggregate, no camera frames, identifiers, or trajectories need be published for neighbors

to benefit; realizing that guarantee end to end is an open problem (Section 6).

Minimal fleets at bottlenecks. The usual objection to vehicle-based control is adoption: no consumer rollout reaches meaningful market share quickly. But market share is the wrong unit. Congestion is not uniform; it is manufactured at bottlenecks, at merges, lane drops, and tunnels, and the network’s worst behavior is set by the inflow those few places receive. The relevant metric is presence in a specific traffic stream at a specific place, which a single fleet operator controls today. Consider the Holland Tunnel, which carries Interstate 78 under the Hudson through four lanes and roughly 10^5 vehicles per day [21]. Every eastbound trip is funneled through approach corridors totaling about 25 feeder lanes, and since adding lanes under the river is impossible, the only lever is the inflow. Ramp metering is exactly this idea, but it requires owning the ramp; a self-regulating car can be introduced one vehicle at a time. The sparse-actuator result suggests one controlled vehicle per roughly twenty damp waves in its neighborhood [27], so one self-regulating vehicle in the platoon window of each feeder lane, about 25 present at a time, is the scale at which the tunnel’s inflow becomes controllable. A single rideshare, parcel, or telematics-insured fleet already has that ambient presence at peak, and the operator capturing the benefit, predictable trip times through the bottleneck, is the one paying the actuation cost. The predecessor points the same way: it modulated speed only at bottleneck segments during peak inflow, and its mixed-traffic ablation improves steadily with the compliant fraction rather than requiring full adoption [2]. The aggregate signal cooperates with this deployment shape: the number of peers a vehicle hears scales with the very density it watches. Near critical density, a 150 m window on a three-lane highway holds roughly 22 vehicles, about one cooperating peer at 5% penetration, and several near jam density. The honest floor: near free flow at very low penetration a vehicle often hears nothing and falls back to local-only control. Where on this penetration–density curve the aggregate begins to change outcomes, and whether bottleneck seeding beats uniform adoption at equal fleet size, are central empirical questions (Sections 4 and 6).

3 Architecture: Minimal Changes to the Vehicle

If minimal fleets are the deployment model, the per-vehicle cost of joining one must be small, so we state the architecture as a bill of changes: what is added to an existing car, and what is not touched. Four functions are added, sensing, an ego-centric digital twin, a learned advisory policy, and a safety and etiquette filter, plus a display; nothing else changes. There is no connection to throttle, brake, steering,

or the CAN control plane; the OBD-II port, on every US car since 1996, is read only, supplying ego speed; no persistent identity leaves the vehicle; and no backend sits in the real-time loop. Training may use centralized data and simulation, but the runtime loop stays on the vehicle, and the kit installs like a dashcam.

Sensing and fusion. Each vehicle estimates its immediate scene from onboard sensors: ego speed, acceleration, lane, headway, leader and follower gaps, relative speeds, local density, queue estimate, and distance to a merge. Aggregate summaries, when heard, add peer-derived state over a bounded radius. The combined observation is deliberately not a global state but an ego-local one, with a few semantic hints about what lies outside direct perception. The deployment stack mirrors the simulator’s observation contract rather than inventing a separate API, in our prototype a 39-field vector vendored verbatim from the training code and covered by bit-equality tests, and logs every field with provenance (measured, derived, fallback-neutral, static, or sim-parity). Ego speed, for instance, is measured from GPS, held during short dropouts, and can be cross-checked against on-vehicle odometry. Leader gap and closing speed come from forward tracks, density from forward counts, rear-lane fields from neutral fallbacks until a rear camera is added, and nearby-vehicle fields from peer summaries when enabled. That provenance map doubles as an observation-quality signal for both the policy and the safety layer.

The ego-centric digital twin. At runtime, each vehicle maintains a lightweight, continuously updated model of its local road state: the ego vehicle, nearby vehicles, lane structure, merge pressure, and short-horizon congestion risk. This is the digital twin. It is a compact tracker, not a city-scale simulator running in the car, and its job is observability. It lets the controller reason about partially observed density and downstream pressure instead of reacting only to the closest leader, and it lets the safety layer test an advisory against a short-horizon prediction before it is shown. The twin also records when each aggregate was last refreshed and from which source; this is what lets the policy distinguish “I see no cooperating vehicles” from “the radio is stale”, and lets the safety layer reject advisories that depend on low-confidence state.

Centralized training, decentralized execution. The learning structure is centralized training with decentralized execution (CTDE). During training, a simulator or a logged fleet can expose global state and counterfactual rollouts; during execution, the policy receives only onboard sensing and local aggregate fields, so no coordinator enters the runtime dependency chain. This division of labor, learning and coordination in the cloud, sensing and control at the edge, is the system’s central placement decision, and the bridge from the centralized predecessor [2] to a vehicle-native one. Because

a deployed policy must meet unseen merge geometries, demand patterns, and compliance levels, we view model-based reinforcement learning, learning a road-network dynamics model and adapting the policy within it, as the most promising route to cross-topology adaptation at acceptable sample cost [11, 13, 29].

Advisory control. The action space stays conservative: speed bins, headway targets, and cautious lane preferences rather than raw throttle, braking, or aggressive lane-change commands. The public action has four fields: a speed bin ('slow', 'nominal', 'fast'), a headway bin, a cautious lane preference, and a merge mode. Prior work on self-regulating cars shows that discrete speed and headway bins of this kind are sufficient for a learned policy to improve flow and reduce hard braking [2], and coarse bins are also what a human driver can actually follow. For human-in-the-loop deployment, the output is a bounded advisory in the spirit of prior advisory-autonomy work [5, 9, 36]; for automated vehicles, it can be passed to an existing longitudinal controller. In both cases the self-regulation layer nudges the vehicle toward network-friendly behavior without claiming full authority over the car.

Safety and etiquette filter. Every advisory passes through a local filter before it is shown or executed. The filter enforces posted speed limits, acceleration and deceleration bounds, minimum front and rear gaps, target-lane time-to-collision margins, a lane-change dwell time, a cap on lane changes per kilometer, and a limit on the braking an advisory may impose on target-lane followers. It also blocks low-speed behavior in uncongested conditions and synchronized all-lane slow-downs unless downstream congestion justifies them. Blocked or modified actions are surfaced through stable diagnostics: safety-masked action, etiquette block, follower-disruption block, and external override. This layer is what allows a network objective to be exposed safely inside mixed human traffic, and it makes the executed action, rather than the raw policy proposal, the quantity that training and evaluation measure.

Degraded operation. The edge design follows one rule: stale data is worse than missing data. If GPS goes stale, the observation builder holds the last valid speed for a short window rather than reporting a stop; if no peers are heard, cooperation fields revert to explicit neutral values rather than silent zeros; if a display or logger stalls, the pipeline drops outputs rather than queuing an advisory that no longer matches the scene. These modes do not make the system complete, but they turn common in-car failures into explicit states rather than hidden dependencies.

Deployment tiers. The same architecture admits three hardware realizations that differ only in which observation fields are measured rather than neutral, a difference the provenance map makes legible to the policy. *Tier 0* is an app: a

modern phone already carries the GPS, inertial sensing, and compute for the observation builder and a two-layer policy, with the camera optional. The advisory rail for this tier operates at scale today: mobile telematics platforms such as CMT and Samsara deliver real-time driving feedback through phone apps and lightweight tags, CMT alone measuring tens of millions of drivers daily [4, 25]. The predecessor argued that this makes app-level integration a credible path from learned speed modulation to system-level coordination without infrastructure changes [2]; our design inherits that path and removes its runtime dependency, shrinking the cloud to training, updates, and offline analytics. *Tier 1* is a retrofit kit, the prototype of Section 4: a windshield camera, a USB GPS receiver, an optional OBD-II dongle as the preferred ego-speed source, an embedded board whose developer kit retails for roughly \$250 [19], and any display. *Tier 2* is native: a vehicle with automated longitudinal control takes the filtered advisory as a set-point for its existing controller. A fleet can mix tiers freely, which makes the 25-vehicle deployment of Section 2 an operational decision rather than a hardware program.

4 Prototype and Measurements

The evidence for tier 1 is a system that exists. We built the full capture-to-advisory stack on a Jetson Orin Nano developer kit, trained a policy in simulation on a desktop machine, deployed it to the device, and validated the loop end to end on simulated drives: prerecorded dashcam footage and a scripted GPS profile pushed through the live pipeline in real time, not an in-vehicle road test. The numbers below establish edge feasibility and behavioral correctness of the deployed loop; they are deliberately not a traffic-flow result, and we close with what would be.

Prototype stack. The board runs JetPack 6 with TensorRT 10.3. Detection is a YOLOv8n engine in TensorRT FP16 invoked directly through the CUDA runtime [22], partly forced by the platform's CPU-only stock PyTorch. Tracking is a SORT-style associator; distance comes from pinhole geometry with vehicle-width priors rather than a dense depth network, a latency-first choice that is swappable if accuracy becomes binding [37]. The observation builder implements the vendored 39-field contract of Section 3; the policy is a two-layer network mirrored in NumPy on the CPU at 0.5 ms per decision. Ego speed comes from a u-blox USB GPS receiver with hold-on-stale handling (an OBD-II feed is planned as the primary source); a dashboard renders the advisory, and every field is logged with provenance.

Policy training. The deployed policy was trained with PPO in a microscopic simulator: a shared policy controlling two vehicles among eight heterogeneous simulated humans on a ring, 3,000 updates of 256 steps, about ninety minutes

Table 1: Measured prototype performance on a Jetson Orin Nano developer kit. Scripted drives replay real dashcam footage with a scripted GPS profile through the live detector-to-advisory stack, paced at 30 Hz. These numbers ground edge feasibility, not traffic-control effectiveness.

Component	Measured point	Notes
YOLOv8n TensorRT FP16 [22]	17.7 ms p50 (3.9 ms GPU)	Forward vehicle detections
Tracking + distance	1.1 ms p50	Leader gap, closing speed
Observation + policy	0.9 ms p50	Advisory proposal
End-to-end compute	19.8 / 20.2 ms p50/p95	Unpaced bench, 48.5 FPS
Scripted cruise drive	24.2 / 25.0 ms p50/p95	8,383 ticks, 29.9 Hz
Scripted braking drive	20.1 / 21.9 ms p50/p95	Leader closes 42 m to 5 m
Ego-speed error	0.121 m/s RMSE	Includes 4 s GPS dropout
Perception coverage	100% / 86.8%	Cruise / braking scenario
Aggregate beacon	2 Hz, 2 s TTL, 150 m	Tens of bytes per message

per seed on one desktop CPU. Five seeds were trained, one of them under sensing noise matched to the deployment measurements (100 m range, 1.5 m position noise, 0.15 m/s speed noise). One training lesson transfers: with a dense speed reward, a one-step crash cost is invisible and the greedy policy collapses to a constant “fast”; an explicit crash penalty restored gap-conditioned behavior. In simulation the learned policy matches, but does not beat, a hand-designed cooperative-smoothing controller on flow (mean speeds 11.8–12.6 versus 11.1 m/s), while an uncontrolled stream collapses into stop-and-go under the same demand. The claim is the behavior’s shape and edge feasibility, not a simulation leaderboard.

Measured performance. Table 1 summarizes the deployed loop. The compute path runs at a 19.8 ms median unpaced and sustains 29.9 Hz on paced drives, with p95 latencies of 21.9–25.0 ms, an order of magnitude inside our 200 ms budget. Ego-speed error over a 280 s drive was 0.121 m/s RMSE, at the injected noise floor and inclusive of a scripted 4 s GPS dropout absorbed by hold-on-stale. The trained policy passed all five pipeline gates (latency, tick rate, GPS freshness, ego-speed error, and perception coverage) on both scenarios. Its advisory behavior, measured post hoc rather than gated, was gap-conditioned: with a leader within 60 m it recommended the nominal bin on all 8,108 such ticks, on open road it recommended fast on 99.6% of ticks (the few exceptions had a

leader tracked just beyond 60 m), and it held nominal through 97% of the low-speed ticks (ego below 12 m/s) of a hard-braking approach. Its advisory churn was 31 head switches per minute against roughly 170 for a random-weights policy, low enough for a human to follow. Offline replay reproduced all 902 decisions of the braking scenario tick for tick, which makes the safety filter auditable after the fact.

Limitations. The dashcam footage is borrowed, so absolute distances carry a factor-of-two scale uncertainty from an assumed field of view; lane assignment and the sign of closing speed are invariant to it, and one-time calibration of the mounted camera removes it. Rear and cooperation fields are currently neutral, leaving mean observation missingness at 0.477 until a rear camera and live peers are added. Advisories still flicker briefly when a leader track is first acquired near a bin boundary; a small hysteresis would damp this. The driver-facing speed readout currently mirrors the raw decode rather than the filter’s gap-aware cap, a display-layer gap we are closing, and an on-road run with a live camera is future work.

Toward a pilot. We plan to build from edge feasibility toward a traffic-flow result in stages: hardware-in-the-loop replay of many vehicles through this stack under swept message loss, delay, sensing range, and compliance; a microscopic-simulation study that replaces the predecessor’s global observation with the local-plus-aggregate contract of Section 2, comparing local-only, local-plus-aggregate, and oracle global-state controllers under the same safety layer; a closed-course study of whether human drivers can follow the advisories without uncomfortable braking; and, as the endpoint, a Holland-Tunnel-shaped corridor study seeding a single instrumented fleet on the feeder lanes of Section 2. The same pilot would exercise the model-based training of Section 3.

5 Related Work

Sparse AVs as traffic actuators. Stern et al. showed a single controlled vehicle can damp stop-and-go waves on a ring track [27], establishing the sparse-actuator premise. Flow and related mixed-autonomy RL work show that learned controllers can improve simulated traffic at low AV penetration [17, 34, 35]. CIRCLES/MegaVanderTest is the strongest field evidence that controlled vehicles can affect open-road traffic, but with purpose-built vehicles and a centralized speed planner over cellular connectivity [6]. The self-regulating-cars formulation learned network-level speed modulation and quantified its benefit, with a centralized observer and cloud broadcast [2]. Our focus is the complement: ordinary vehicles, edge-executed control, and a runtime that needs no coordinator.

Connected-vehicle messaging. Deployed connected-vehicle standards have each car broadcast its own position, speed, and heading for collision avoidance at up to ten hertz: the

SAE J2735 Basic Safety Message over DSRC and the ETSI Cooperative Awareness Message over C-V2X [7, 16, 24]. Within the same family, the Collective Perception Service shares per-object detection records [8], and probe-vehicle systems compute neighborhood traffic state by uplinking per-vehicle traces to a backend that aggregates centrally [12, 26]. The same radios carry our messages; the difference we rely on is that our unit of exchange is a fixed-size, receiver-side neighborhood aggregate, with no sender addressed, retained, or reconstructable and no backend in the loop, which keeps the privacy surface small and the message size independent of scene complexity.

Infrastructure-centric traffic control. Ramp metering, variable speed limits, adaptive signal control, and freeway management systems remain the dominant deployed tools [1, 20, 32, 33]. They are powerful where sensors and actuators exist, but they depend on fixed infrastructure and centralized operational visibility. Self-regulating cars instead treat vehicles as mobile actuators that can be introduced incrementally.

Advisory systems. Human-compatible advisory systems show how network objectives translate into driver-facing recommendations [5, 9, 36]. Early deployment will be mixed: some vehicles automated, some showing dashboard advisories, many ignoring the system. Mobile telematics platforms show that the driver-facing rail itself already operates at the scale of tens of millions of drivers [4, 25]. The open systems question is therefore not only control optimality, but compliance, incentives, and fairness under partial adoption.

Edge AI and networked sensing. Embedded perception models and Jetson-class hardware make on-vehicle traffic-state estimation increasingly practical [19, 22, 37]. Cloud video analytics demonstrates the value of centralized visual processing but also the cost of moving raw data off the edge [14, 15]. Our communication model takes the opposite position: infer locally, exchange only semantic aggregate state, and keep safety-critical control on the vehicle.

6 Discussion: Open Networking Questions

A self-regulating fleet is, at bottom, a networked system: edge nodes deciding locally, a low-rate substrate among them, and a cloud kept off the control path. Taking its deployment seriously turns the hardest remaining questions into networking ones.

How little communication is enough, and can it be private? Section 2 fixes semantics, not sizes. We need principled positions on rate, precision, aggregation radius, freshness, and trust: how stale a signal may be before it misleads, how coarse before it stops helping, and how a receiver should weight summaries of unknown provenance. Privacy sharpens the question. Our contract keeps the consumed unit anonymous, but a naive realization still has each sender

emit its own position, as a Basic Safety Message already does; the open problem is to produce the aggregate without any raw per-vehicle trajectory leaving the vehicle, through in-network or gossip aggregation, or differentially private counts that stay useful to a flow regulator yet carry no linkable identity.

What belongs in the cloud, and how late may it be?

The runtime loop is fully local and offline-capable by construction; what remains is training, periodic model updates, and fleet analytics over decimated uplinks of a kilobyte per vehicle-second or less. Delay tolerance is thus a design property rather than a hope: an advisory never waits on backhaul. The open question is how much slowly arriving global context, incidents or corridor demand, is worth folding back into the local policy, and on what cadence.

How robust is the system to partial deployment?

Evaluation should sweep penetration, compliance, message loss, sensing noise, and demand bursts across a topology ladder from rings to merges to inverted trees, directly compare bottleneck seeding against uniform adoption at equal fleet size, and report throughput, jam fraction, and delay alongside hard braking, collisions, and follower delay, so a network gain is never bought by making human traffic less safe.

How should safety be verified? A local policy can optimize a network objective and still be unacceptable if it surprises human drivers. The safety layer should be specified, tested, and audited separately from the learned controller, with explicit checks for stale state, follower disruption, low-speed obstruction, and unsafe merges. This separation can become a systems requirement: the executed-action diagnostics of Section 3 report how often a suggestion was masked, blocked, or overridden, giving continuous evidence that a network objective is being achieved safely.

Who benefits, and who decides? A voluntary advisory system can shift delay across drivers, lanes, or vehicle classes. A deployable system therefore needs measurement and incentives that make the network benefit visible and its distribution accountable, especially in the fleet-seeded model, where one operator acts and every driver rides the smoother flow.

As vehicles become networked computers, congestion control need not remain a roadside service: the controller fits in coarse bins, the radio is already deployed, the compute costs a few hundred dollars, and a single commercial fleet is enough presence at a real bottleneck. The open work is to give each vehicle just enough local traffic state to help the network while it acts independently, safely, privately, and without a cloud coordinator; we think this is a productive problem space for the networking community.

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